

AUTOMOTIVE SERVICE MANUAL

Brake System - Front Disc Brakes

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Vehicle Type: Standard Passenger Vehicles

SECTION 1: INTRODUCTION

This section covers the inspection, removal, and installation of front disc brake pads and related components. Follow all procedures carefully to ensure proper brake operation and vehicle safety.

WARNING:

Always wear safety glasses and gloves when working on brake systems. Brake dust may contain asbestos fibers - use proper respiratory protection.

CAUTION:

Do not use compressed air to clean brake components as it may spread hazardous dust particles.

NOTE:

It is recommended to replace brake pads in pairs (both sides) to ensure even wear and consistent braking performance.

SECTION 2: INSPECTION PROCEDURES

2.1 Visual Inspection

Park the vehicle on a level surface and engage the parking brake. Loosen the front wheel lug nuts while the vehicle is on the ground.

Step 1: Lift the Vehicle

Use a hydraulic jack to lift the front of the vehicle. Place jack stands under the designated lift points as indicated in Figure 1.

Step 2: Remove the Wheel

Remove the lug nuts and take off the front wheel. Set aside in a safe location.

Step 3: Inspect Brake Pads

Locate the brake caliper and inspect the brake pad lining thickness. Minimum acceptable thickness is 3mm. See Table 1 for specifications.

Step 4: Check Rotor Condition

Examine the brake rotor surface for scoring, grooves, or visible wear marks. Measure rotor thickness using a micrometer. Replace if below minimum specification.

Table 1: Brake Pad Specifications

- Original Pad Thickness: 12mm
- Minimum Service Thickness: 3mm
- Recommended Replacement Interval: 30,000-50,000 miles
- Pad Material: Ceramic (standard) or Semi-Metallic (performance)

SECTION 3: REMOVAL PROCEDURE

3.1 Caliper Removal

To access the brake pads, the caliper must be removed. Support the caliper with wire - do not let it hang from the brake hose.

Step 1: Loosen Caliper Bolts

Using a socket wrench, loosen the caliper guide bolts. Apply penetrating oil if bolts are corroded.

Step 2: Remove Caliper

Remove the bolts and carefully slide the caliper off the rotor. Hang it securely using wire or a suitable hanger.

Step 3: Remove Old Brake Pads

Slide the brake pads out of the caliper bracket. Note the orientation of the wear indicators for proper installation.

SECTION 4: INSTALLATION PROCEDURE

4.1 Installing New Pads

Apply brake grease to the back of the new pads in the areas indicated in Figure 2. Do not apply grease to the friction surface.

Step 1: Compress Piston

Use a C-clamp or piston compression tool to push the caliper piston back into its bore. This makes room for the new, thicker brake pads.

Step 2: Install New Pads

Place the new brake pads into the caliper bracket, ensuring the wear indicator is positioned correctly.

Step 3: Reinstall Caliper

Slide the caliper back over the rotor and pads. Install the guide bolts and tighten to the manufacturer's specified torque (typically 25-35 ft-lbs).

Step 4: Reinstall Wheel

Place the wheel back on the hub and hand-tighten the lug nuts. Lower the vehicle and tighten lug nuts in a star pattern to 100 ft-lbs.

SECTION 5: POST-INSTALLATION CHECKS

WARNING:

After installation, you **MUST** test drive the vehicle in a safe area to verify brake function before returning to normal driving.

Test Procedure:

1. Start the engine and let it idle for 1 minute.
2. Apply light brake pressure to seat the pads - repeat 5-10 times.

3. Drive at low speed (20 mph) and test brake responsiveness.
4. Gradually increase speed and test braking at various speeds.
5. Listen for unusual noises such as squealing or grinding.

NOTE:

New brake pads may require 200-300 miles to fully break in. Expect slightly longer stopping distances during this period.

SECTION 6: TROUBLESHOOTING

Common Issues:

- Squealing brakes: Check pad installation, may need brake grease on back of pads
- Brake pedal pulsation: Rotor may be warped - check runout with dial indicator
- Reduced braking power: Check for air in brake lines or worn pads
- Brake drag: Check caliper slides are moving freely, apply appropriate grease

APPENDIX A: TORQUE SPECIFICATIONS

- Wheel Lug Nuts: 100 ft-lbs (135 Nm)
- Caliper Guide Bolts: 25-35 ft-lbs (34-47 Nm)
- Caliper Bleeder Valve: 10-15 ft-lbs (14-20 Nm)

APPENDIX B: REQUIRED TOOLS

- Hydraulic jack and jack stands
- Socket set (metric and SAE)
- Torque wrench
- Brake bleeder kit
- C-clamp or piston compression tool
- Brake grease
- Wire for hanging caliper

APPENDIX C: FIGURE REFERENCES

Figure 1: Jack stand lift points - See vehicle owner's manual

Figure 2: Brake pad grease application points

Figure 3: Brake system hydraulic layout

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